

TERM: Orphanet

Plain English

Database of rare diseases.

Why it matters

Used to identify diseases and associated genes.

Input

Disease names.

Output

Disease IDs, genes, prevalence info.

Used in pipeline

Disease discovery stage.

How to verify manually

Search disease directly on orphanet website.

Useful links

<https://www.orpha.net/>

What I still don't understand

Difference between Orpha ID and OMIM ID.

Your ROGDI case study has a major identity error still spread through the repo.

A live UniProt check says:

ROGDI → Q9GZN7, ROGDI_HUMAN, "Protein rogdi homolog", 287 aa

But lots of your files still say:

ROGDI / GMPR2 → Q9P2T1, 348 aa, GMP reductase 2

That older mapping is wrong for ROGDI. It poisons the deep dive, fixture, tests, generated report

So your next move is not "build more pipeline". Your next move is:

Stop expanding. Stabilise the scientific truth layer first.

How to approach the whole project

Think of nano-rare-gt as having two separate layers:

1. Science truth layer

This is the manually verified knowledge base:

- What disease is this?
- What gene actually causes it?
- Which UniProt accession is correct?
- What does the protein actually do?
- Which tissues matter?
- Is AAV feasible?
- What are the delivery paradoxes?
- What evidence supports each claim?

2. Software automation layer

This is the code that turns verified facts into repeatable outputs:

- Pydantic models
- database storage
- API clients
- scoring engine
- report renderer
- CLI

Right now, the software layer is healthier than the science layer. The code passes tests, but some

Your project should now proceed like this:

Phase A: ROGDI manual truth audit

Before more coding, manually rebuild the ROGDI case study from source links.

Phase B: Correct the project's golden fixture

The fixture should become the single truth source for ROGDI. Tests should verify the corrected fa

Phase C: Regenerate reports

Only after the fixture is corrected should the report and protocol be trusted.

Phase D: Build pipeline modules

Once ROGDI is scientifically clean, then add Orphanet, OMIM, UniProt, vector, homology, and scori